

Extremality shift of rotating black holes at finite Gauss–Bonnet coupling

A research project carried out with AI agents

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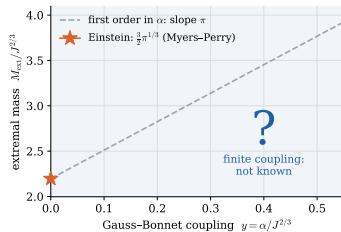
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With AI assistance: **Claude Code** (Anthropic; Claude Opus 5, Sonnet 5)
and **Codex** (OpenAI; GPT-5)

Gravitational Waves and AI-Assisted Research — M. Zaldarriaga & N. Mirón Granese
Departamento de Física, FCEN, Universidad de Buenos Aires · 2 October 2026 ·
repo github.com/ddblanco/BHs

The question: how far does Gauss–Bonnet move the spin bound?



M87*, EHT (2019)

General Relativity (in 5D, two equal spins $J_1 = J_2 = J$)

$$\text{Kerr (4D): } M^2 \geq |J| \quad \text{Myers-Perry (5D): } M \geq \frac{3}{2} \pi^{1/3} J^{2/3}$$

extremal case: bound saturates $\rightarrow T = 0$

Add Gauss–Bonnet term, coupling α :

$$I = \frac{1}{16\pi G} \int \sqrt{-g} (R + \alpha \mathcal{L}_{\text{GB}})$$

schematically $R \sim g \partial^2 g + \partial g \partial g$

$$\mathcal{L}_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$$

Known to first order in α (arXiv:2009.00015):

$$M_{\text{ext}} = \frac{3}{2} \pi^{1/3} J^{2/3} + \pi \alpha + \mathcal{O}(\alpha^2)$$

Asked: $\partial M_{\text{ext}}/\partial \alpha|_J$ at **finite** α ?

Hint: numerical rotating BHs (arXiv:1010.0860)

Main result: the bound tightens, but not linearly

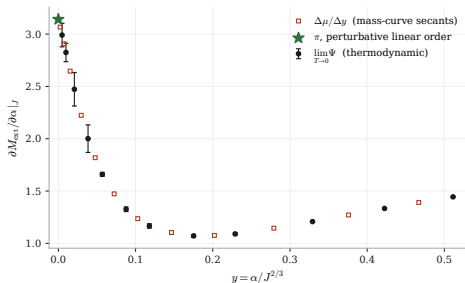
extended first law

$$dM = T dS + 2\Omega_H dJ + \Psi d\alpha$$

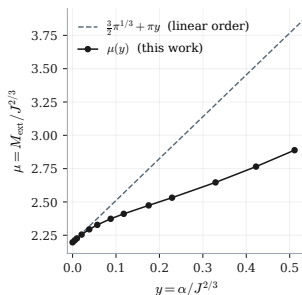
at $T = 0$, fixed J

slope of the extremal mass

$$\Rightarrow \left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = \Psi_{\text{ext}}$$



slope $\partial M_{\text{ext}} / \partial \alpha|_J$ (result)



extremal mass

> 0

bound tightens

$\pi \rightarrow 1.07 \rightarrow 1.44$

not monotonic

+31%

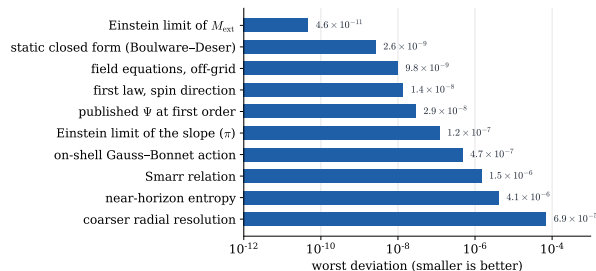
growth of M_{ext}

2.3×

first-order shift vs.
measured

356 numerical solutions at 13 couplings; the $T = 0$ values are extrapolated from $TJ^{1/3} \approx 3 \times 10^{-6}$.

Validation: how do we know it is right?



worst deviation of each independent check

1. Known answers first

published family (arXiv:1010.0860) to 2×10^{-4} ; static closed form; π at $\alpha \rightarrow 0$

2. Independent routes to Ψ

Smarr, from one solution's charges:

$$\Psi = \frac{2M - 3TS - 6\Omega_H J}{2\alpha}$$

on-shell Gauss-Bonnet action:

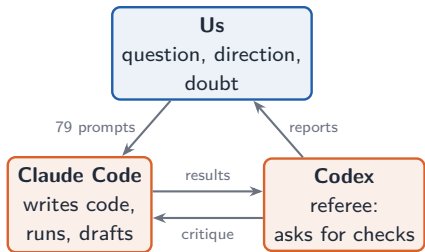
$$\Psi = -\frac{1}{16\pi} \int_{\Sigma} \sqrt{-g} \mathcal{L}_{\text{GB}}$$

neither uses our linear solve: agree to $\sim 10^{-6}$

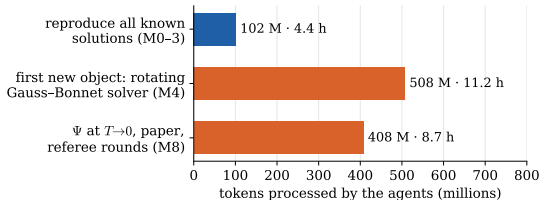
3. Largest uncertainty

$T \rightarrow 0$ is an extrapolation (the error bars)

What we learned working with agents



A check the referee asked for **failed**: our error bars were too small (two ways of computing the slope disagreed) → we enlarged them.



36.8 h

agent time over 13 days

1.42 billion

tokens processed

Lesson: the cost is not writing code, it is verifying it.

Ask for checks that **can fail**, and let one agent referee the other.

Thank you for your attention

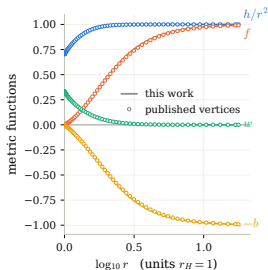
Matías and Nahuel, thanks for the AI course

A1 · Numerical method: solving the field equations

Equal-spin ansatz $\Rightarrow$ ODEs in r for b, f, h, w :

$$ds^2 = -b dt^2 + \frac{dr^2}{f} + r^2 \left[d\theta^2 + \sin^2\theta \cos^2\theta (d\varphi_1 - d\varphi_2)^2 \right. \\ \left. + h \left[\sin^2\theta d\varphi_1 + \cos^2\theta d\varphi_2 - w dt \right]^2 \right]$$

horizon: $b = f = 0$, $w = \Omega_H$; infinity: $b, f, h/r^2 \rightarrow 1$, $w \rightarrow 0$



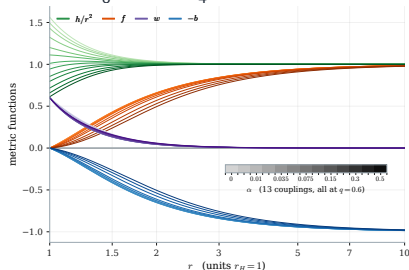
Discretise: $\chi = 1 - r_H/r$, Chebyshev–Lobatto, $N = 64$: $R_N(u; \alpha, r_H, \Omega_H) = 0$ ($4N$ eqs.)

Solve: damped Newton; continuation from Myers–Perry

Accept: $G_{\mu\nu} + \alpha H_{\mu\nu} < 10^{-8}$ at 51 off-grid points

Read off: $b \simeq 1 + U/r^2$, $w \simeq W/r^4 \Rightarrow$

$$M = -\frac{3\pi}{8} U, J = \frac{\pi}{4} W$$



A2 · Numerical method: measuring Ψ and reaching $T = 0$

Ψ from derivatives at fixed (r_H, Ω_H)

$$\Psi = \left. \frac{\partial M}{\partial \alpha} \right|_{S,J} = M_\alpha - T S_\alpha - 2\Omega_H J_\alpha$$

Derivatives without finite differences

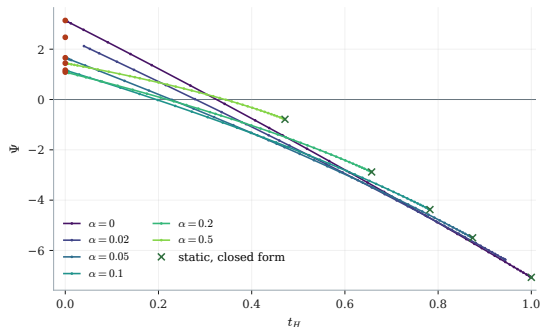
$$\underbrace{\frac{\partial R_N}{\partial u}}_{\text{Newton Jacobian}} \frac{du}{d\alpha} = - \frac{\partial R_N}{\partial \alpha}$$

one linear solve, no step size (implicit function theorem)

Get cold, then extrapolate

cubic fit through the 6 coldest states;

coldest $TJ^{1/3} \approx 3 \times 10^{-6}$



Ψ against temperature; red dots: the $T = 0$ values (the slope)

A2b · Two other routes to Ψ

1. Smarr relation (scaling)

weights: $M, \alpha \rightarrow 2$, $S, J \rightarrow 3$; Euler's theorem:

$$2M = 3TS + 6\Omega_H J + 2\alpha\Psi$$
$$\Rightarrow \Psi = \frac{2M - 3TS - 6\Omega_H J}{2\alpha}$$

charges of **one** solution: no Jacobian, no derivative
agrees to 1.45×10^{-6} (313 states); useless as $\alpha \rightarrow 0$

2. On-shell Gauss–Bonnet action

Gibbs potential = on-shell action per unit time; at fixed (T, Ω_H) , α enters explicitly only through $\mathcal{L}_{\text{GB}}$ (field equations remove the rest):

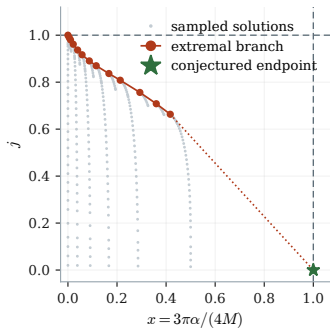
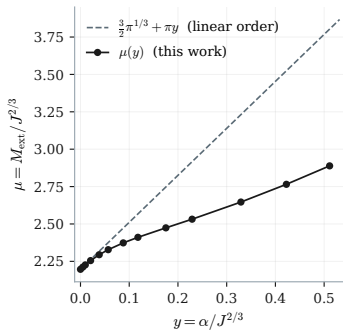
$$\Psi = -\frac{1}{16\pi} \int_{\Sigma} \sqrt{-g} \mathcal{L}_{\text{GB}}$$

curvature of the solution only: no charge extraction either

agrees to 4.67×10^{-7} (43 states, away from extremality)

Both use the same numerical solutions; neither uses the linear solve of A2.

A3 · Physical meaning of the result



Spin bound

$\Psi_{\text{ext}} > 0 \Leftrightarrow j_{\text{ext}}$ falls:
less spin per unit mass

Slope minimum

$$3\omega = \mu - y\mu'$$

= fastest extremal horizon,
 $\omega = \Omega_H J^{1/3}$

Why it rises again

$M > \frac{3\pi}{4}\alpha$ forces
 $\mu' \rightarrow \frac{3\pi}{4} \approx 2.36$

A4 · How it started: the first prompts

“El proyecto consiste en utilizar agentes de IA para implementar códigos numéricos que permitan obtener soluciones de agujeros negros rotantes en dimensión $D > 4$ [...] reproducir soluciones numéricas ya encontradas en la literatura. [...] el paso siguiente sería emprender la búsqueda de nuevas soluciones.”

Prompt 1 — 4 Sept, 19:16, to Codex · then “Fijate que el proyecto realmente siga las indicaciones propuestas por el docente del curso” (3), “ejecutar con subagentes” (5), “continuemos con el hito 3”, “dale, avanza” · 79 human prompts, 4–28 Sept · Claude Code builds ↔ Codex referees

```
ruta-de-trabajo-inicial.txt 5 Sept 2026
52 ## Hito 4 – EGB rotante asintóticamente plano
53
54 *Salida mínima exitosa del proyecto:* familia parcial, pero validada, de
55 soluciones D=5 con momentos angulares iguales que reproduzca resultados de
56 Brihaye, Kleihaus, Kunz y Radu (2010).
57
58 - Continuar primero en velocidad angular y luego en alpha.
59 - Comparar magnitudes adimensionales y curvas publicadas.
60 - Comprobar regularidad, cargas, temperatura, entropía y primera ley.
61 - Documentar zonas donde el método pierde convergencia sin interpretarlas de
62 inmediato como fronteras físicas.
63 ... (Hito 5 – método híbrido asistido por IA)
64 ## Hito 6 – Exploración nueva
65
66 *Salida ambiciosa:* evidencia numérica reproducible sobre una extensión no
67 cubierta por los benchmarks.
68
69 Sólo se seleccionará después del Hito 4. Candidatos: otra densidad de Lovelock,
70 otra condición asintótica o una región no explorada del espacio de parámetros.
71 El spin desigual o un ansatz de cohomogeneidad 2 se consideran proyectos de
72 segunda generación por requerir PDEs.
```

first roadmap, 5 Sept. Hito 6 became Ψ ; Hito 8 ($T \rightarrow 0$, paper, referees) was not in the plan.

4.1.3 Thermodynamical properties

Considering the thermodynamics of these solutions, the EGB black holes should satisfy the first law of thermodynamics

$$dE = T_H dS + 2\Omega_H dJ. \quad (4.3)$$

One may regard the parameters S, J as a complete set of extensive parameters for the mass-energy $E(S, J)$ and define the intensive parameters conjugate to them. These quantities are the temperature and the angular velocities. Also, for $\alpha \neq 0$, these solutions do not appear to satisfy any simple Smarr relation¹⁴.

¹⁴The Smarr relation in Einstein gravity is $E = 3(T_H S + 2\Omega_H J)/2$. The recent work [43] proposes a Smarr relation for static black holes in Lovelock gravity. It would be interesting to extend the formalism in [43] to the case of spinning solutions.

arXiv:1010.0860, §4.1.3 and footnote 14.

Hito 6, α as a thermodynamic variable:

$$2M = 3TS + 6\Omega_H J + 2\alpha\Psi$$

A5 · What it cost

36.8 h

active agent time, 13 days

1.42 billion

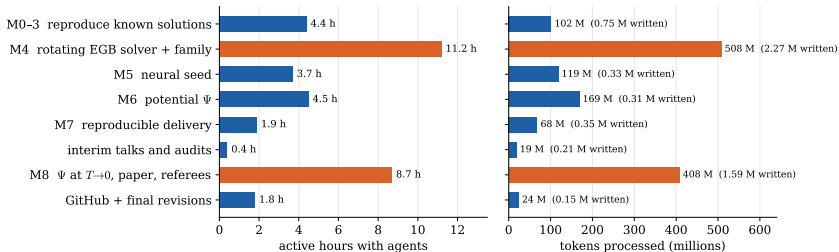
tokens processed
(99% re-read context)

5.9 million

tokens written (≈ 24 MB)

79

human prompts



The first **new** object cost $5\times$ the tokens of reproducing everything known.

Not counted: human time, background numerical runs.